The Benefits of OpenSSH

The SSH service is very widely used in open source infrastructure set-ups. Due to its small footprint on the network, as well as the ease of installation and maintenance, SSH replaces many remote shells in data centres today. OpenSSH is a very famous flavour of this protocol, and this article talks about its benefits and the challenges associated with it.

OpenSSH is free Secure Shell software that provides great services in terms of protocol-based connectivity as well as security. It replaces almost all legacy applications such as Telnet, rlogin, etc. It is important to understand how the protocol works, and the bells and whistles provided in terms of features. Please refer to Figure 1, which shows the protocol stack forming the OpenSSH protocol services. For those who know how Telnet works, it is easy to understand the working operations of OpenSSH. Similar to Telnet, it runs as a daemon service on Linux servers, while the client uses an SSH client utility such as *putty* to connect to the server. SSH is available on Windows as well as UNIX platforms, and is widely used on Linux infrastructures. By default, it uses TCP port 22 for communication. However, unlike the Telnet protocol, OpenSSH is primarily used to ensure data security and, to that end, it uses cryptography to authenticate the client and the server, and also for data transfer purposes. This ensures confidentiality and integrity of the data that flows on the wire.

Its communication has three basic steps—the client-server handshake, authentication and secure data exchange. During the handshake phase, both the sides exchange information about the OpenSSH protocol version and the cipher algorithms they support (which are typically the combinations of asymmetric and symmetric encryption, as well as hashing algorithms and compression algorithms). Unlike SSL, in this protocol, the server sends the first data block to the client.

As for authentication, the server is authenticated using the host key, whereas the client typically stores the key fingerprint at some predefined location and validates it later in the process. Please see Table 1, which shows the supported client authentication methods.